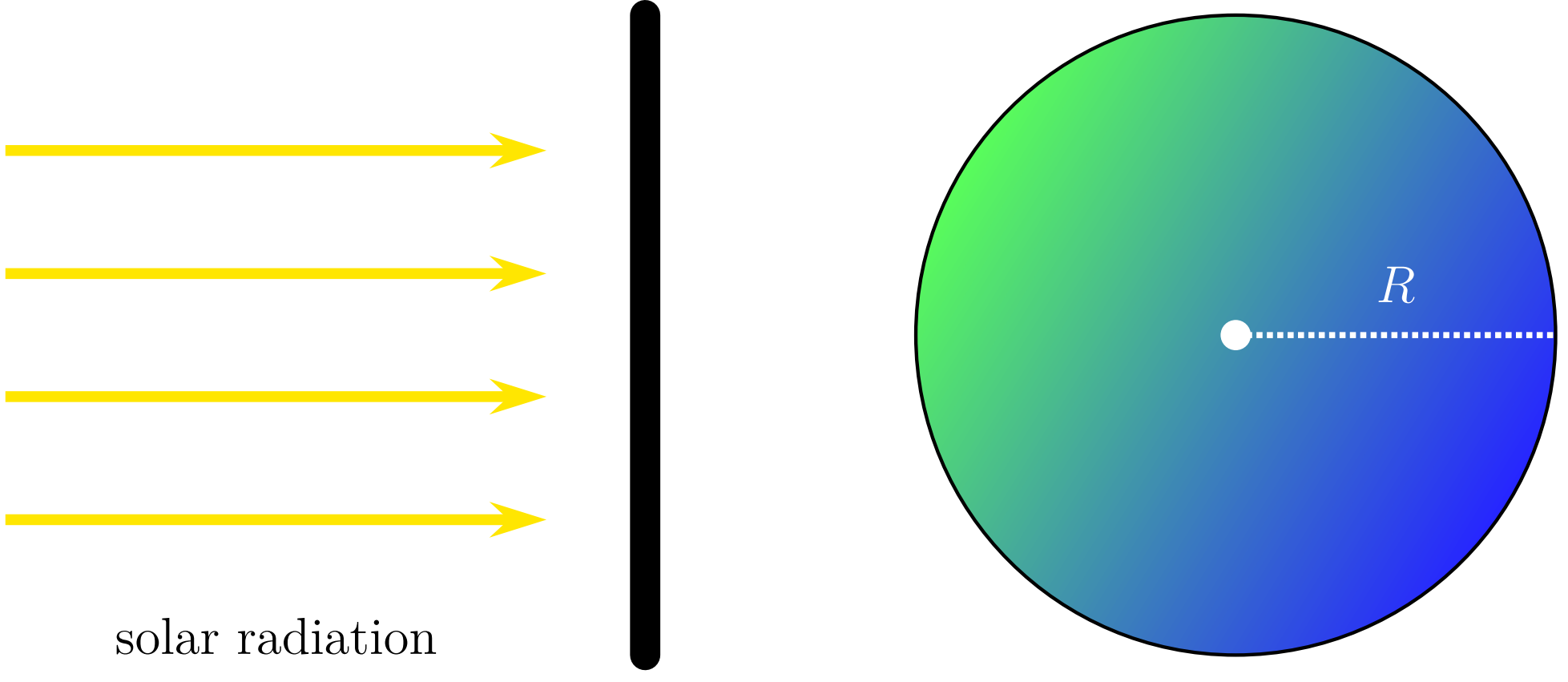
R
solar radiation
disc of area $4\pi R^2$
surface area of the sphere $4\pi R^2$